

Engineering Mathematics III

ENGE702

Lecture 16: Laplace Transforms

Laplace Transforms

The power of Laplace transforms comes from their application in solving differential equations. Specifically it can help with step functions, and when functions have been 'shifted', or are piecewise functions.

In the first part of the lecture we will look at Laplace transforms of shifts in the time domain.

Laplace Transforms

An important function here is called the **Heaviside function**. It is defined as:

$$u(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases}$$

Then we can shift this function to the right by a distance a , so:

$$u(t - a) = \begin{cases} 0 & \text{if } t < a \\ 1 & \text{if } t \geq a \end{cases}$$

This becomes useful if we want to 'switch on' a function at a certain point a :

$$f(t)u(t - a) = \begin{cases} 0 & \text{if } t < a \\ f(t) & \text{if } t \geq a \end{cases}$$

Laplace Transforms

We can combine step functions to create a 'pulse' that turns on at time a , and off again at time b , where $a < b$.

This is done by simply taking $f(t) = u(t - a) - u(t - b)$.

Laplace Transforms

We can find the Laplace transform of a function that has had the Heaviside function applied using the **second translation theorem** or t -shift theorem:

$$\mathcal{L}(f(t-a)u(t-a)) = e^{-as}\mathcal{L}(f(t))$$

There is a technical note that we need to take care of: we have $f(t-a)$ on the left and $f(t)$ on the right.

Laplace Transforms

Example: Find $\mathcal{L}(f(t))$, where $f(t)$ is given by:

$$f(t) = \begin{cases} 0 & \text{if } t < \pi \\ -\sin(t) & \text{if } t \geq \pi \end{cases}$$

Laplace Transforms

Exercise: Find $\mathcal{L}(f(t))$, where $f(t)$ is given by:

$$f(t) = \begin{cases} 0 & \text{if } t < 1 \\ t^3 - 3t^2 + 3t - 1 & \text{if } t \geq 1 \end{cases}$$

Text Book References

Chapters 6.2, 6.3, 6.4